

Week 12 Software Lesson: Sample Size and Power in R

Goals

The goal of this software lesson is to learn how to

- estimate required sample sizes for comparisons of means and proportions, and
- create power curves and tables.

Required Packages

You will need to install and load the following packages in R for this software lesson:

- `pwr`

INTRODUCTION

An important step in doing research is determining the appropriate sample size for a given study. When figuring this out, you must consider Type I error, power, and cost. For any given statistical test, and assuming parameters for the underlying population distribution, the following four values have a close relationship with one another:

1. Sample size
2. Minimum Clinically Important Difference (MCID)
3. Significance level, α
4. Power

If you know three of the four, you can determine the fourth quantity.

This software lesson will focus on prospective sample size estimation when comparing two groups, that is, estimating a required sample size given a desired significance level, power, and MCID.

METHODS

In this software lesson, we will first learn to calculate a sample size from the significance level, power, and minimum clinical significant difference in group means. Then we will create tables and curves to show the power for multiple sample sizes.

We will then learn how to calculate sample size and create power tables and curves for the difference in group proportions.

DIFFERENCES IN GROUP MEANS

Estimating sample sizes

You can use the `pwr.t.test()` function from the `{pwr}` package to estimate sample sizes when comparing group means. The necessary arguments for this function are

- `d=`, which is the effect size statistic *Cohen's d*, $d = \frac{|\mu_1 - \mu_2|}{\sigma}$, to calculate MCID; a small effect corresponds to roughly $d=0.20$, a medium effect to $d=0.50$, and a large effect to $d=0.80$.
- `sig.level=`, which is the significance, or alpha, level of the test; also known as the Type I error rate.
- `power=`, which is the power of the test.
- `type=`, which is the type of *t*-test ("two.sample" when comparing two means, "one.sample" when have single mean, "paired" when have paired data).

There is also an option for specifying the alternative hypothesis, if it is different from two-sided; the default is set at two-sided tests.

For example, the following code estimates the required sample size needed in each group to detect a small effect of 0.20 when you set the significance level at 0.05 and power at 0.90.

```
library(pwr)

twogrpmean.n <- pwr.t.test(d = 0.20, sig.level = 0.05, power = 0.90, type = "two.sample")

twogrpmean.n
```

The output will provide you with an estimated sample size (`n =`) for each group as well as each of the inputs that you provided in the `pwr.t.test()` function (i.e., `d =`, `sig.level =`, `power =`, and `alternative =`). Note that the recommended sample size is the recommended size for *each* group.

Interpretation of Results

Using a 5% significance level, 1054 patients (527 in each group) are required to have a 90% chance of detecting an effect size *d* of 0.20 between two group means.

Creating a power curve for one MCID

To view the power curve, which shows values of power for many possible sample size values, simply use the `plot()` function on the `pwr` object.

```
plot(twogrpmean.n)
```

Your plot will look like this:

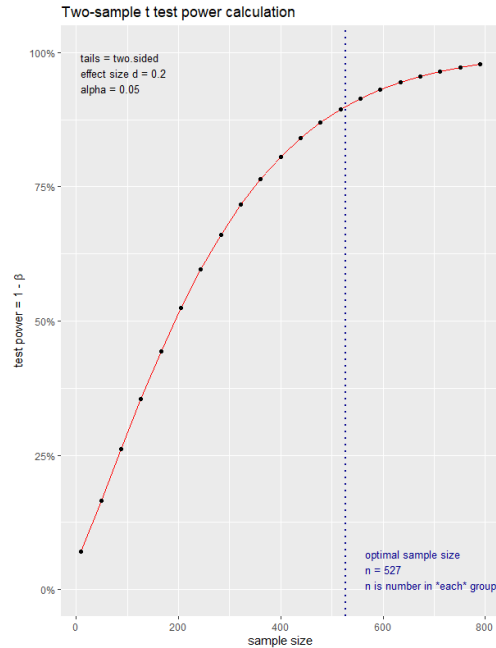


Figure 1: Power curve for comparing two means for one MCID

Creating a power table for one MCID

To generate a power table, you need to use `pwr.t.test()` to estimate power values for many possible sample sizes. The easiest way to do this is to specify candidate sample size values by creating an array containing each candidate value using the `seq()` function and the `from=`, `to=`, `by=` arguments. For example, to generate the list: 0.5, 0.6, 0.7, 0.8, 0.9, type `seq(from=0.5, to=0.9, by=0.1)`.

Then, use this array object in the sample size argument (`n=`) in `pwr.t.test()` to estimate the power values for many possible sample sizes.

```
candidate.n <- seq(from = 25, to = 1000, by = 25) #creates a list 25, 50, 75, 100, ...
twogrpmean.power <- pwr.t.test(d = 0.2, sig.level = 0.05,
                               n = candidate.n, type = "two.sample")
```

The output is an object that stores many pieces of information:

- `$n` for an array of specified sample sizes
- `$d` for the specified effect size
- `$sig.level` for the specified significance level
- `$power` for an array of calculated power values
- `$alternative` for the specified alternative hypothesis
- `$note`

To view a table with just sample sizes and estimated powers, create a new data frame, extracting the the sample sizes (`$n`) and the power (`$power`) from the `pwr` object.

```
twogrpmean.pwrtable <- data.frame(n = twogrpmean.power$n,
                                  power = twogrpmean.power$power)

twogrpmean.pwrtable
```

Creating power curves to compare two different MCIDs

To plot two different power curves on the same plot, you need to use the array of sample size values again, but use it to create two different power curves, and then plot both of them.

Our example above used $d = 0.2$. Let's create data for $d = 0.5$ and $d = 0.8$, which we will call `twogrpmean.power.medium` and `twogrpmean.power.large` respectively.

```
# creating list of sample sizes
n.values <- seq(from = 5, to = 100, by = 5)

# getting power for each sample size in list
twogrpmean.power.medium <- pwr.t.test(d = 0.5, sig.level = 0.05,
                                     n = n.values, type = "two.sample")

twogrpmean.power.large <- pwr.t.test(d = 0.8, sig.level = 0.05,
                                    n = n.values, type = "two.sample")

# plotting the first curve
plot(n.values, twogrpmean.power.medium$power, type = "b",
     xlab = "sample size", ylab = "power", ylim=c(0,1))
lines(n.values, twogrpmean.power.medium$power, type = "l")

# adding the points with dashed lines of the second curve to plot
points(n.values, twogrpmean.power.large$power, type = "b", col=2, lty=2, pch = 2)

# adding a legend for each curve
legend("bottomright", c("d = 0.5", "d = 0.8"), lty = c(1,2),
      pch = c(1,2), col = c(1,2), cex=1.5)
```

Your plot will look like this:

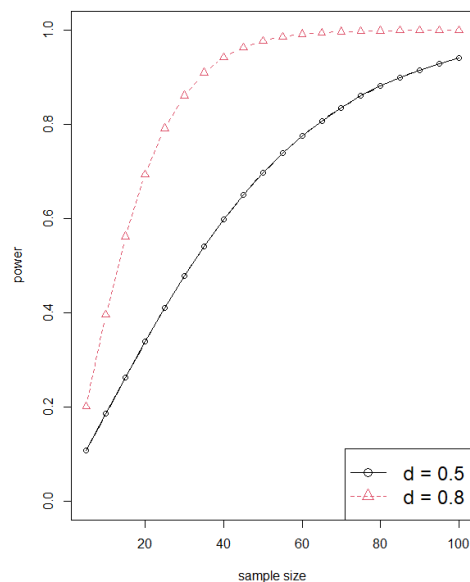


Figure 2: Power curves for comparing two means for two MCIDs

DIFFERENCES IN GROUP PROPORTIONS

Estimating sample sizes

Estimating sample sizes when comparing group proportions is very similar to when comparing group means. Instead of the `pwr.t.test()` function, use the `pwr.2p.test()` function from the `{pwr}` package. Furthermore, instead of specifying the effect size directly, use the `ES.h()` function to estimate *Cohen's h* effect size for two proportions.

For example, the following code estimates the required sample size needed in each group when the two proportions are 0.85 and 0.70 and you set the significance level at 0.05 and power at 0.90.

```
twogrpp.n <- pwr.2p.test(h = ES.h(p1 = 0.85, p2 = 0.70), sig.level = 0.05, power = 0.90)

twogrpp.n
```

The output will provide you with an estimated sample size (`n =`) for each group as well as each of the inputs that you provided in the `pwr.2p.test()` function (i.e., `h =`, `sig.level =`, `power =`, and `alternative =`). Note that the recommended sample size is the recommended size for *each* group.

Interpretation of Results

Using a 5% significance level, 318 patients (159 in each group) are required to have a 90% chance of detecting an effect size *h* of 0.36 between two group proportions.

Creating a power curve for one MCID

To view the power curve, simply use the `plot()` function on the `pwr` object.

```
plot(twogrpp.n)
```

Your plot will look like this:

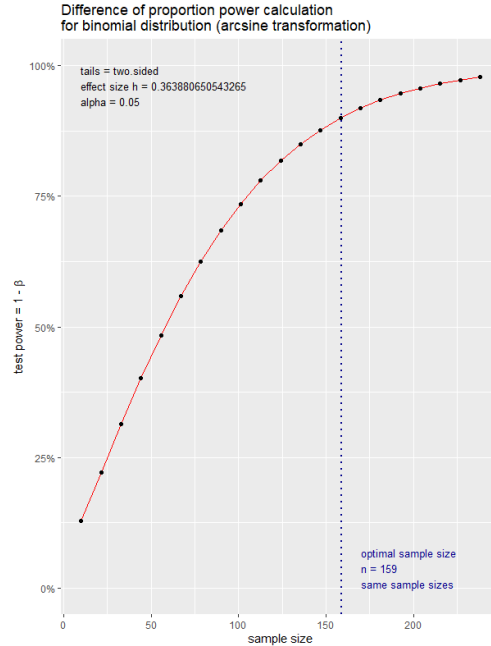


Figure 3: Power curves for comparing two proportions for one MCIDs

Creating a power table for one MCID

Creating the power table is exactly the same as in the two means situation (i.e., create an array of candidate sample size values and put those into the `pwr.2p.test()` function to calculate the power values for those sample sizes. Then create a new data frame that has only the sample sizes (`$n`) and the power (`$power`) from the `pwr` object).

```
candidate.n <- seq(20, 500, by = 20) #creates a list 20, 40, 60, ...

twogrpp.power <- pwr.2p.test(h = ES.h(p1 = 0.85, p2 = 0.70),
                             sig.level = 0.05, n = candidate.n)

twogrpp.pwrtable <- data.frame(n = twogrpp.power$n,
                               power = twogrpp.power$power)

twogrpp.pwrtable
```

Creating power curves and tables to compare two different MCIDs

To plot two different power curves on the same plot, you need to use the array of sample size values again, but use it to create two different power curves, and then plot both of them.

```
candidate.n <- seq(from = 20, to = 500, by = 20)

twogrpp.power70 <- pwr.2p.test(h = ES.h(p1 = 0.85, p2 = 0.70),
                               sig.level = 0.05, n = candidate.n)

twogrpp.power75 <- pwr.2p.test(h = ES.h(p1 = 0.85, p2 = 0.75),
                               sig.level = 0.05, n = candidate.n)
```

```

# Tables
twogrpp.power70.table <- data.frame(n = twogrpp.power70$n,
                                     power = twogrpp.power70$power)
twogrpp.power70.table

twogrpp.power75.table <- data.frame(n = twogrpp.power75$n,
                                     power = twogrpp.power75$power)
twogrpp.power75.table

# Plots
n<-twogrpp.power70$n
p70<- twogrpp.power70$power
p75<- twogrpp.power75$power

plot(n, p70, type = "b", xlab = "sample size", ylab = "power", ylim=c(0,1))

points(n, p75, type = "b", col=2, lty=2, pch = 2)

legend("bottomright", c("p2=0.70", "p2=0.75"), lty = c(1,2),
      pch = c(1,2), col = c(1, 2), cex=1.5)

```

Your plot will look like this:

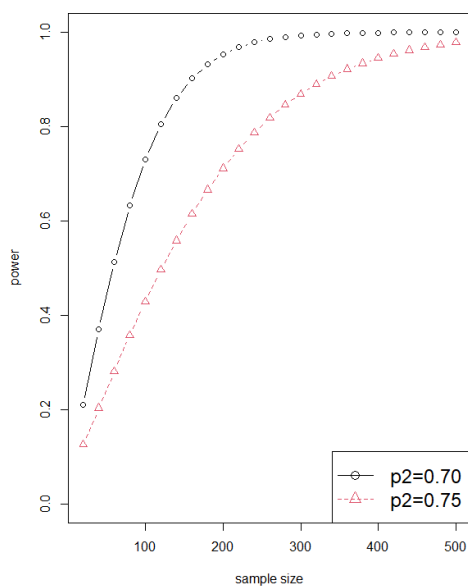


Figure 4: Power curves for comparing two proportions for one MCIDs

Additional Resources

For additional R resources on sample size estimation and power analysis, see:

- Power Analysis [Quick-R].
- Sample Size Estimation and Power Analysis [RPubs].
- Sample Size Calculation with R [Dr. Williamson, UND].